10.1 Standard OB Transport Operations 10-2
10.1.1 Patient Assessment 10-2
10.1.2 Preparation for Transport 10-2
10.1.3 During Patient Transport 10-3
10.2 General Guidelines for the Postpartum Patient 10-6
10.2.1 Patient Assessment and Management Prior to Transport 10-6
10.2.2 During Patient Transport 10-6
10.3 Vaginal Bleeding, Placenta Previa, Abruption 10-7
10.3.1 Patient Assessment and Management Prior to Transport 10-7
10.3.2 During Patient Transport 10-7
10.4 Postpartum Hemorrhage 10-8
10.4.1 Additional Transport Considerations 10-8
10.5 Premature Rupture of Membranes (PROM) 10-9
10.5.1 Patient Assessment and Management Prior to Transport 10-9
10.5.2 During Patient Transport 10-9
10.5.3 Additional Transport Considerations 10-10
10.6 Preterm/Active Labor 10-11
10.6.1 Patient Assessment and Management Prior to Transport 10-11
10.6.2 Treatment During Transport 10-11
10.6.3 Additional Transport Considerations 10-12
10.7 Hypertension, Preeclampsia, Eclampsia, HELLP Syndrome 10-13
10.7.1 Patient Assessment and Management Prior to Transport 10-13
10.7.2 During Patient Transport 10-13
10.7.3 Pregnancy Induced Hypertension (PIH) or Pre-eclampsia 10-14
10.7.4 Additional Transport Considerations 10-16
10.8 Thyroid Storm 10-17
10.8.1 Patient Assessment 10-17
10.8.2 During Patient Transport 10-17
10.9 Electronic Fetal Monitoring 10-18
10.9.1 NICHD Three-Tier Fetal Heart Rate Interpretation System 10-18
10.9.2 NICHD Fetal Monitoring Terminology 10-19

10.1 Standard OB Transport Operations

10.1.1 Patient Assessment

Obtain patient history including past medical, surgical and obstetrical history. Include complications with prior pregnancies.

Obstetrical History and Assessment

Obtain current information including current pregnancy complications, obstetrical testing or ultrasound results and reason for transport.

- Obtain a handover report
- Baseline vital signs (blood pressure, pulse, respiratory rate, temperature, and SpO2)
- Obtain serum glucose PRN and repeat as needed
- Uterine tone and activity
- Fetal heart rate (FHR) or fetal heart pattern
- Status of membranes
- Presenting part
- Pelvic exam if indicated (cervical dilation, effacement, and station).

Electronic Fetal Monitor

If patient is on an EFM, assess, categorize and document the recent electronic fetal monitor strip activity using the NICHD 3 tier system.

10.1.2 Preparation for Transport

- Establish and maintain an IV with LR at 50-100 mL/hr
- Ensure a combined fluid intake total of 125-150 mL/hr taking into consideration IV medication infusions
- Maintain the patient in a left or right lateral position for uterine displacement
- Position cot/sled restraint straps below the uterus

UNCONTROLLED WHEN DOWNLOADED OR PRINTED

10.1.3 During Patient Transport

Maternal Monitoring and Assessment

- Monitor maternal SpO2, respiratory rate, heart rate, blood pressure at least every 15 minutes
- Assess uterine activity and tone at least every 15 minutes

Electronic Fetal Monitoring

Monitor the fetus with a EFM during transport if the following criteria are met:

- The patient is 23 weeks gestation or greater
- As indicated by specific fetal condition
- If the patient is < 23 weeks gestation the EFM may be used to monitor FHR

NOTE: **FHR may be more difficult to capture at gestational ages < 23 weeks**

Assessment

- Assess and record fetal heart tones or fetal heart rate pattern and fetal movement at least every 15 minutes for patients 23 weeks gestation or greater
- Assess and record fetal heart rate at the referring and receiving facility if the patient is 18-22 weeks gestation.
- Assess and record any changes in maternal or fetal condition

No EFM

If unable to use EFM

- Assess and record fetal heart tones via Doppler
- Ask the patient if they are feeling fetal movment

Oxygen

Administer supplemental oxygen PRN to maintain oxygen saturation 96% to 99% using the following criteria:

- By nasal cannula (2-4 L/min) if maternal oxygen saturation is 96% to 99% and fetal heart rate monitor interpretation shows Category I [Normal] FHR tracing.

  NOTE: **Category I tracing includes all of the following: baseline 110-160 bpm, moderate variability, no late or variable decelerations. Accelerations and early decelerations may be present or absent.**
- By non-rebreather mask (10-12 L/min) if maternal oxygen saturation is less than 96% or fetal heart rate monitoring interpretation does not show a Category I FHR tracing [Normal] or if fetal status is unknown.

Other Treatment

- Administer fluid bolus PRN (Refer to Preterm Labor, Active Labor and Fetal Stabilization/Resuscitation Protocol Guidelines)

Labor Pain or Severe Non-labor Pain

NOTICE **Use analgesia for labor pain that is unrelieved with tocolysis or non-pharmacological approaches.**

Fentanyl

Obstetrical

- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

  NOTE: **For pain related to headache or labor contractions**

  NOTE: **Do not administer opioid analgesics if delivery is possible within 1 hour of administration**

Analgesia

Acetaminophen

- 650-1000 mg PO

Fentanyl

- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

  NOTE: **For pain related to headache or labor contractions**

  NOTE: **Do not administer opioid analgesics if delivery is possible within 1 hour of administration**

Nausea or Vomiting

Ondansetron

Obstetrical

- 4 mg IV over 2-5 min
- May repeat once in 30 min PRN
- Do not exceed 8 mg in 8 hours

Promethazine

Obstetrical

- 12.5 mg diluted in 10 mL IV over 2 min
- Alternately 25 mg deep IM
- May repeat q 4–6 hours PRN

NOTE: **Ondansetron or promethazine may be administered concurrently with magnesium sulfate to prevent nausea**

Anxiety

Midazolam

Obstetrical

- 2–5 mg IV over 1-2 min, IM, IN
- Max single dose 5 mg
- May give 2 mg, observe for effect, and then give an additional dose if indicated

Lorazepam

Obstetrical

- 0.5–1 mg IV

Magnesium Toxicity

Magnesium is used to treat several obstetrical conditions which puts patients at risk for symptomatic hypermagnesemia. Consider and treat hypermagnesemia with calcium chloride or calcium gluconate if the patient looses deep tendon reflexes, suffers muscle weakness or has impaired respiration.

Calcium Chloride

Obstetric

- 0.5–1 gm IV over 3 min
- May repeat x 1 if symptoms do not resolve

Calcium Gluconate

Obstetric

- 0.5-1 gm IV over 3 min
- May repeat x 1 if symptoms do not resolve

Additional Transport Considerations

- Reassure the patient, and explain any procedures and the transport process to the patient and family,
- Notify receiving labor and delivery unit if support personnel and/or equipment are needed emergently on arrival.
- If birth is imminent at sending facility, assist facility staff in preparing for birth, notify Communications if neonatal specialty team is needed.

Medical Control

Contact the medical director, medical control, or the receiving Maternal Fetal Medicine specialist or OB Hospitalist (MFM/OB) for any of the following:

- Maternal or fetal condition has deteriorated from initial report
- If maternal or fetal condition is unstable and transport may be contraindicated due to advancing dilatation, labor progress, or possibility of delivery during transport
- Maternal or fetal condition requires prompt interventions available at the sending facility
- If required by local practice or requested by the transport Medical Director

UNCONTROLLED WHEN DOWNLOADED OR PRINTED

10.2 General Guidelines for the Postpartum Patient

10.2.1 Patient Assessment and Management Prior to Transport

Obtain an additional postpartum OB history and assessment that includes the following:

- Length of labor
- Induction agents used
- Time of delivery. Include time of placenta delivery and whether it was spontaneous or manual

  NOTE: **The placenta should be brought to receiving hospital if possible**
- Mode of delivery (vaginal or cesarean section). Include APGAR scores and infant weight if available
- Estimated blood loss (EBL)
- Fetal or maternal indications for delivery (preeclampsia, eclampsia, PROM, PTL, term, etc)
- Fundal height, location and amount of lochia, perineal lacerations or episiotomy
- Results of any labs, urinalysis, or other diagnostic tests performed
- Obtain accurate I&O's
- Medications given PTA, including time, route and dosage

10.2.2 During Patient Transport

Stimulation of Uterine Contractions

Initiate or continue pharmacologic stimulation of uterine contractions **after** delivery of the placenta

- Perform fundal massage until uterus is firmly contracted and every 10 minutes thereafter
- Note the presence or absence of clots and the flow of bleeding accompanied by the fundal massage.

Oxytocin

- 30 units in 500 mL (0.06 units/mL) of NS or LR
- 500–1000 mL/hr
- Titrate infusion rate to sustain uterine contraction and control uterine atony

  NOTE: **If no IV is established, administer 10 units IM**

Analgesia for Uterine Contractions

Acetaminophen

- 650-1000 mg PO

Fentanyl

- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

  NOTE: **For pain related to headache or labor contractions**

  NOTE: **Do not administer opioid analgesics if delivery is possible within 1 hour of administration**

10.3 Vaginal Bleeding, Placenta Previa, Abruption

10.3.1 Patient Assessment and Management Prior to Transport

- Additional history: all medications taken at home, history of illicit drug use, bleeding disorders, recent ASA or NSAID use and prior ultrasound results regarding placental location
- Onset time of bleeding, precipitating events, EBL, color of blood (pink, bright red, dark red), presence of clots
- Results of current ultrasound regarding placental location, abruption or presence of vasa previa
- Note the results of any diagnostic labs: blood type, CBC, PT, PTT, fibrinogen, etc.
- Note time of PTA medications given, including route and dosage. Note the time any blood products, including RhoGAM, were given, including patient response
- Obtain accurate I&Os

10.3.2 During Patient Transport

- Assess and record maternal vital signs at least every 15 minutes, including patient temperature if blood products given
- Consider fluid replacement (LR or NS): infuse 3 mL fluid for every 1 mL of estimated blood loss
- IV fluids to be maintained at a minimum of 125-150 mL/hr
- Place a second 18 ga or greater IV line for active bleeding or suspected abruption

⚠ CAUTION **Avoid digital vaginal exam unless placenta previa has been ruled out.**

Additional Transport Considerations

Tocolysis

Administer magnesium sulfate to achieve fewer than one contraction every 10 minutes.

Magnesium

Obstetric

- 6 gm IV over 20–30 min initial 1 time loading bolus
- 3–4 gm/hr continuous infusion following initial loading bolus
- Titrate infusion every 45-60 min PRN to suppress contractions

Online Medical Control

Contact the medical director, medical control, or the receiving MFM/OB for the following:

- Concerning hemodynamic changes in patient
- Significant changes to the fetal monitor strip such as a category I normal strip changes to a category II indeterminate strip.
- Guidance regarding maternal/fetal stability for transport

10.4 Postpartum Hemorrhage

Postpartum Hemorrhage is blood loss greater than 500 mL after spontaneous vaginal delivery, or greater than 1000 mL after cesarean section.

- Consistent with regional practice, treat cardiovascular instability and hypotension with IV fluids, blood products and TXA.
- Establish a second IV site PRN
- Perform fundal massage to assist with uterine contractions
- Place a urinary catheter to empty the bladder and facilitate uterine contraction PRN

Oxytocin

- 30 units in 500 mL (0.06 units/mL) of NS or LR
- 500–1000 mL/hr
- Titrate infusion rate to sustain uterine contraction and control uterine atony

  NOTE: **If no IV is established administer 10 units IM**

Tranexamic Acid

- 1 gm over 10 minutes
- After 30 minutes, reassess for bleeding
- 1 gm over 10 minutes if bleeding continues

  NOTE: **Administer the first dose within 3 hours of the bleeding onset.**

  NOTE: **Ensure there is no evidence of retained placenta, full bladder or prolapsed uterus prior to administration.**

  NOTE: **The second dose may be administered if bleeding recurs within 24 hours of the initial bleeding event.**

  NOTE: **Medical control should be contacted as soon as practicable.**

NOTICE **Oxytocin and Tranexamic Acid may be given per guidelines. Other medications listed for postpartum hemorrhage may only be given with online medical control.**

10.4.1 Additional Transport Considerations

- Assess fundal height and amount of lochia with each vital sign or at least every 15 min

Medical Control

Contact the medical director, medical control, or the receiving MFM/OB for the following:

- Concerning hemodynamic changes in the patient
- Guidance regarding maternal stability for transport

10.5 Premature Rupture of Membranes (PROM)

10.5.1 Patient Assessment and Management Prior to Transport

Refer to General Guidelines for the Obstetrical Transport and obtain a targeted OB history and assessment that includes the following:

- Presence of a cervical cerclage and date of placement
- Previous administration of antenatal steroids and dates administered
- History of rapid labor
- Time ROM occurred and the color and odor of leaking fluid
- Time of last vaginal exam or sterile speculum exam and any changes over the course of the hospitalization

Diagnostic Tests

Review all available diagnostic tests or labs specifically inquiring about:

- Ferning
- Amnisure (or equivalent)
- Amniotic fluid index (AFI)
- Group B beta hemolytic strep (GBS) screen

Medications

Medications PTA

Reconcile all medications given prior to arrival including:

- Route
- Dose
- Response

Home Medications

Reconcile all home medications specifically inquiring about:

- Nifedipine
- Progesterone
- Ibuprofen
- Terbutaline
- Antibiotics

Vaginal Exam

Perform or have the sending staff repeat a sterile vaginal exam or sterile speculum exam if indicated:

- Urge to push
- Urge to have a bowel movement
- Active labor
- Bedside to bedside transport time >30 min

10.5.2 During Patient Transport

- Transport with hips elevated if cervical dilation is greater than 3-4 cm
- A fluid bolus of 250-500 mL may be given for dehydration, possible infection, or hypoxemia
- Refer to Preterm / Active Labor Guideline and initiate tocolysis if contractions are present or if cervix is dilated greater than 3 cm

10.5.3 Additional Transport Considerations

- Contact the medical director, medical control, or the receiving MFM/OB if advancing dilatation with possibility of delivery during transport occurs or there are significant changes to the fetal monitor strip e.g. Category I normal strip changes to a category II indeterminate strip.
- Significant changes to the fetal monitor strip (i.e.: changing from a Category I [normal] strip to a category II [indeterminate] or category III [abnormal] strip) occurs

10.6 Preterm/Active Labor

10.6.1 Patient Assessment and Management Prior to Transport

Refer to General Guidelines for the Obstetrical Transport and obtain a targeted OB history and assessment that includes the following:

- All home medications specifically inquiring about:
  - Nifedipine
  - Progesterone (17P)
  - Ibuprofen
  - Terbutaline
- Presence of a cervical cerclage and date of placement
- Previous administration of antenatal steroids and dates administered
- History of rapid labor
- Time of last vaginal exam or sterile speculum exam and any changes over the course of the hospitalization
- Review and note all available diagnostic tests or labs specifically inquiring about
  - Fetal Fibronectin (fFN) screen
  - Group B beta hemolytic strep (GBS) screen
  - Cervical length via ultrasound
- Note time of PTA medications given, including route, dosage and response. Obtain accurate I & Os
- Perform or have the sending staff repeat a sterile vaginal exam or sterile speculum exam, as needed or indicated: urge to push, urge to have a bowel movement, active labor, long transport time (>30 min).

10.6.2 Treatment During Transport

Transport with hips elevated if cervical dilation is greater than 3-4 cm.

An empiric fluid bolus of 250-500 mL may be given for suspected dehydration or infection

Tocolysis

Administer tocolytics PRN to achieve fewer than one contraction every 10 minutes.

NOTICE **If contractions are more frequent than one every 6 minutes, administer terbutaline to provide fast acting tocolysis while magnesium sulfate is being prepared for administration.**

Terbutaline—Tocolysis

- 0.25 mg SQ
- Repeat every 20–30 min PRN

NOTE: **Tocolysis for contractions more frequent than one every 6 minutes**

If terbutaline is contraindicated, ketorolac may be administered.

Ketorolac

- 30 mg IV over 30–60 seconds
- May repeat 1 x after 60 min

Treatment During Transport

Magnesium sulfate

- 6 gm IV over 20–30 min initial 1 time loading bolus
- 3–4 gm/hr continuous infusion following initial loading bolus
- Titrate infusion every 45-60 min PRN up to 4 gm/hr to suppress contractions

NOTICE **If less than 6 gm was previously administered and effective tocolysis has not been achieved, a bolus to bring the total Magnesium Sulfate bolus to 6 gm should be administered followed by the maintenance infusion**

NOTE: **Magnesium is contraindicated if nifedipine has been administered within the last hour**

Analgesia

Labor pain unrelieved with tocolysis may be treated with opioid analgesics

Fentanyl

Obstetrical

- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

  NOTE: **For pain related to headache or labor contractions**

  NOTE: **Do not administer opioid analgesics if delivery is possible within 1 hour of administration**

10.6.3 Additional Transport Considerations

Document contraction frequency, contraction duration, and response to tocolytics.

Contact the medical director, medical control, or the receiving MFM/OB if any of the following occur:

- Advancing dilatation or labor with the possibility of delivery during transport
- Significant changes to the fetal monitor strip (i.e.: changing from a Category I [normal] strip to a category II [indeterminate] or category III [abnormal] strip) occurs

10.7 Hypertension, Preeclampsia, Eclampsia, HELLP Syndrome

10.7.1 Patient Assessment and Management Prior to Transport

Refer to General Guidelines for the Obstetrical Transport and obtain a targeted OB history and assessment that includes the following:

Home medications

Specifically inquiring about:

- Labetalol
- Methyldopa

History

- Chronic hypertension and typical blood pressure ranges

Assessment

Presence with onset time or absence of the following:

- Headache
- Visual changes and abnormalities
- Upper abdominal or epigastric pain
- Edema
- Shortness of breath
- Hyperreflexia
- Clonus

Vital Signs and Medications

NOTE: **Identify blood pressure upon admission and any changes to blood pressure during hospitalization**

- Review and note all available diagnostic tests or labs
- Note time of PTA medications given, including route, dosage and response.
- Obtain accurate fluid input and output

10.7.2 During Patient Transport

Reassure the patient, reduce external stimuli, provide hearing protection, and use vision protection during transport.

10.7.3 Pregnancy Induced Hypertension (PIH) or Pre-eclampsia

Initiate or continue seizure prophylaxis using magnesium sulfate

Magnesium

Obstetric

- 4–6 gm IV over 20–30 min initial 1 time loading bolus
- 2 gm/hr continuous infusion following initial loading bolus

NOTICE **Increase the dose for patients presenting with 4+ reflexes and sustained clonus, edema up to the hips, headache and visual changes with elevated blood pressure**

NOTICE **Contact medical control (i.e., sending physician, receiving physician, or program medical director) to increase magnesium sulfate dosage.**

Eclampsia

Magnesium

Obstetric

- 6 gm IV over 15-20 min Initial 1 time loading bolus
- 2 gm/hr continuous infusion following the initial loading bolus

NOTE: **If a less than 6 gm loading bolus was given PTA, an additional bolus to bring the total magnesium sulfate bolus to 6 gm should be administered.**

IV Fluid Management

Maintain total fluid intake at 125ml/hr,

NOTE: **Fluid intake may be increased for hypovolemia or fetal intrauterine resuscitation**

Hypertensive Treatment

Administer an antihypertensive if there are 2 systolic BP readings > 160 mmHg or 2 diastolic BP readings > 110 mmHg in 15 minutes time.

- Begin treatment within 10-20 minutes
- Target SBP < 160 mmHg and DBP below 110 mmHg

NOTE: **Use chosen antihypertensive through its administration protocol before switching to an alternative antihypertensive**

Labetalol

- 20 mg IV over 2 minutes
- Repeat 40 mg in 10–20 min PRN
- Repeat 80 mg every 10–20 min PRN to a max of 300 mg

NOTE: **Administer if there are 2 SBP readings > 160 mmHg or 2 DBP readings > 110 mmHg in 15 minutes time**

NOTE: **Contact medical director for dosage guidance if the patient is on a labetalol infusion**

NOTE: **Administer max dose of labetalol prior to administering another antihypertensive**

Hydralazine

- 10 mg IV over 2 min
- Repeat 10 mg IV push over 2 min If target BP is not reached 20 min after first dose
- Repeat 5 mg IV push over 2 min If target BP is not reached 20 min after second dose
- Maximum total dose 30 mg
- Administer to target DBP 90–105 mm/Hg or 20% reduction

Do not administer, or stop administration if:

- The heart rate is greater than 120

NOTE: **Acceptable alternative to labetalol**

Nifedipine

- 10 mg PO
- 20 mg PO if target BP is not reached after 30 minutes
- 20 mg PO if target BP is not reached after second dose
- Max dose 50 mg before switching antihypertensive agents

NOTE: **Repeat BP every 15 minutes**

NOTE: **Administer if there are 2 SBP readings > 160 mmHg or 2 DBP readings > 110 mmHg in 15 minutes time**

NOTE: **Administer max dose of nifedipine prior to administering another antihypertensive.**

UNCONTROLLED WHEN DOWNLOADED OR PRINTED

Eclamptic Emergency

Benzodiazepines

Maintain airway, provide supplemental O2, and administer a benzodiazepine for seizure control (choose one)

Lorazepam

Obstetric

- 2 mg IV, IM, NAS
- Repeat every 5 minutes until the seizure is controlled

Midazolam

Obstetric

- 5 mg IV, IM, NAS

Magnesium

Initiate or continue magnesium sulfate bolus and maintenance IV drip

Magnesium

Obstetric

- 6 gm IV over 15-20 min Initial 1 time loading bolus
- 2 gm/hr continuous infusion following the initial loading bolus

  NOTE: **If a less than 6 gm loading bolus was given PTA, an additional bolus to bring the total magnesium sulfate bolus to 6 gm should be administered.**

Analgesia for Headache

Fentanyl

Obstetrical

- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

  NOTE: **For pain related to headache or labor contractions**

  NOTE: **Do not administer opioid analgesics if delivery is possible within 1 hour of administration**

Acetaminophen

Obstetrical

- 650-1000 mg PO

10.7.4 Additional Transport Considerations

Contact the medical director, medical control, or the receiving MFM/OB the following:

- Patient experiences seizures or major changes in symptoms
- For further guidance regarding antihypertensive treatment
- Changes in the fetal monitoring strip i.e. (changing from a Category I [normal] strip to a Category II [indeterminate] or Category III [abnormal] strip) occur
- Onset of additional complications (active labor, abruption, neurological deterioration, etc.)

10.8 Thyroid Storm

10.8.1 Patient Assessment

Perform a physical assessment including baseline vital signs (blood pressure, pulse, respiratory rate, temperature, and SpO2.

Handover Report

Obtain a handover report and perform an obstetrical history and physical exam to include:

- Home medications
- History of patients thyroid disorder
- History of Graves disease
- History of any possible precipitating events (surgery, infection, HTN, diabetes, etc)
- Note presence or absence of fever, tachycardia, cardiac dysrhythmias, tremors, mental status changes, nausea or vomiting,
- Note results of all labs, urinalysis, ultrasounds or any other diagnostic tests performed
- Obtain accurate I&O's
- Note medications given PTA, including time, route and dosage.
- Contact Medical Direction or discuss with sending physician the administration of Propylthiouracil (PTU)

Propylthiouracil

- 300–600 mg PO loading dose
- 150–200 mg PO every 4–6 hours
- Requires an online physician order

Acetaminophen

- 650-1000 mg PO

10.8.2 During Patient Transport

- Initiate or continue cardiac monitoring
- Maintain fluid/electrolyte replacement
- Initiate or continue cooling measure such as wet blankets
- Place a nasogastric tube if patient unable to swallow

Additional Transport Considerations

Contact the medical director, medical control, or the receiving MFM/OB for the following:

- Any major changes in patient condition
- For guidance regarding further treatment
- If changes in the fetal monitoring strip i.e. (changing from a Category I [normal] strip to a Category II [indeterminate] or Category III [abnormal] strip) occur.

10.9 Electronic Fetal Monitoring

10.9.1 NICHD Three-Tier Fetal Heart Rate Interpretation System

Category I

Category I FHR tracings include all the following:

- Baseline FHR: 110-160 bpm
- Baseline FHR variability: moderate
- Late or variable decelerations: absent
- Early decelerations: present or absent
- Accelerations: present or absent

Category II

Category II FHR tracings include all tracings not categorized as Category I or Category III

Category III

Category III FHR tracings include:

Absent baseline FHR variability and any of the following:

- Late decelerations
- Variable decelerations
- Bradycardia
- Sinusoidal pattern

10.9.2 NICHD Fetal Monitoring Terminology

Variability

Fluctuations in FHR baseline that are irregular in amplitude and frequency

- Measured in a 10-minute window
- The amplitude is measured peak to trough
- There is no distinction between short-term and long-term variability

Definition

- Absent—amplitude undetectable
- Minimal—amplitude 0 to 5 bpm
- Moderate—amplitude 6 to 25 bpm
- Marked—amplitude over 25 bpm

Baseline FHR

Mean beats per minute rounded to nearest 5 beats over a 10-minute interval excluding:

- Accelerations and decelerations
- Periods of marked variability
- Segments that differ by more than 25 bpm

NOTE: **There must be at least two minutes of identifiable baseline in any 10 minute widow or the baseline for that period is considered indeterminate**

Definitions

- Bradycardia—below 110 bpm
- Normal—110 to 160 bpm
- Tachycardia—over 160 bpm

Acceleration

An abrupt increase in the FHR above baseline

- Before 32 weeks of gestation, accelerations should last ≥10 sec and peak ≥10 bpm above baseline
- As of 32 weeks gestation, accelerations should last ≥15 sec and peak ≥15 bpm above baseline

Prolonged Acceleration

- Increase in the FHR that is 15 or more bpm above baseline
- Lasts ≥ 2 min but <10 min

NOTE: **An acceleration of 10 minutes or more is considered a change in baseline**

Late deceleration

A gradual usually symmetrical decrease and return to baseline of the FHR associated with a uterine contraction

- The deceleration is delayed in timing, with the nadir of the deceleration occurring after the peak of the contraction
- Onset of deceleration to nadir ≥ 30 seconds
- The onset, nadir, and recovery usually occur after the onset, peak, and termination of a contraction

Early deceleration

A gradual usually symmetrical decrease and return to baseline of the FHR associated with a uterine contraction.

- The nadir of the FHR and the peak of the contraction occur at the same time.
- Onset of deceleration to nadir ≥ 30 seconds
- The deceleration's onset, nadir, and termination are usually coincident with the onset, peak, and termination of the contraction.

Variable deceleration

An abrupt decrease in FHR below the baseline

- The decrease is ≥15 bpm, lasting ≥15 secs and <2 minutes from onset to return to baseline
- Onset of deceleration to nadir < 30 seconds
- The onset, depth, and duration of variable decelerations commonly vary with successive uterine contractions

Prolonged deceleration

- Decrease in the FHR that is 15 or more bpm below baseline
- Lasts ≥ 2 min but <10 min

NOTE: **A deceleration of 10 minutes or more is considered a change in baseline**

Rate of Change

Changes are measured from the onset of the deceleration/acceleration to its nadir/peak.

- Gradual changes = greater than 30 seconds
- Abrupt changes = less than 30 seconds

References

1. National Institue of Child Health and Human Development Research Planning Workshop. Am J Obstet Gynecol 1997; 177:1385.
2. Macones GA, Hankins GD, Spong CY, et al. The 2008 National Institute of Child Health and Human Development Workshop Report on Electronic Fetal Monitoring: Update on Definitions, Interpretation, and Research Guidelines. Obstet Gynecol 2008; 112:661.

UNCONTROLLED WHEN DOWNLOADED OR PRINTED